Distilling Clinical Reasoning into Small Models

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You can find the full project repository here:
GitHub Repository - Clinical Reasoning Distillation

Abstract

Distilling large language models into smaller, deployable systems is critical for clinical decision support, but reasoning-intensive tasks involve multiple components—decision, reasoning, and structure—that are not equally transferable. We study distillation through the lens of capability decomposition, analyzing how different training objectives shape distinct aspects of model behavior. We show that while objectives induce complementary capabilities, combining them through multi-objective training consistently fails under constrained adaptation.

We address this limitation by shifting to representation-level adaptation, applying LoRA to MLP layers under a decision-oriented objective. This yields the strongest performance observed, matching or exceeding teacher-level decision accuracy while achieving near-perfect detection of unsafe cases at the decision level and improving reasoning-related metrics.

These findings highlight a fundamental limitation of objective-level combination and demonstrate the importance of where adaptation is applied within the model for effective knowledge transfer.

1 Introduction

The integration of artificial intelligence into clinical decision-making presents both significant opportunities and fundamental challenges. While large language models demonstrate strong performance on complex reasoning tasks, their size, computational requirements, and reliance on external infrastructure make them difficult to deploy in real-world clinical environments, where privacy, reliability, and cost constraints are critical. As a result, there is a growing need for smaller, locally deployable models that can match the performance of larger systems, particularly in high-stakes settings such as medical decision support.

Knowledge distillation provides a natural framework for addressing this challenge by transferring behavior from a high-capacity teacher model to a smaller student. However, in reasoning-intensive domains, distillation is not a uniform process. A model's output consists of multiple components—such as the final decision, the supporting reasoning, and the structural format—and these components may not be equally easy to transfer. In particular, while final predictions can often be learned effectively, transferring deeper reasoning capabilities remains challenging.

In this work, we study distillation through the lens of *capability decomposition*. Rather than treating supervision as a single signal, we analyze how different training objectives influence distinct aspects of model behavior, including decision accuracy, safety, and reasoning quality. We first

show that different objectives induce different capabilities, revealing a structured but fragmented landscape of model behavior. We then investigate whether these capabilities can be combined through multi-objective training, and find that, under constrained adaptation, such combinations consistently fail to improve over the best single-objective models.

Motivated by this limitation, we shift focus from objective design to the *representation level* of the model. We show that modifying where adaptation is applied—specifically, moving from attention layers to MLP layers—leads to substantial and consistent improvements across all evaluation metrics. These models achieve the strongest performance observed in this work, including matching or exceeding teacher-level decision accuracy while achieving near-perfect detection of unsafe cases at the decision level and significantly improving reasoning-related metrics.

Our results highlight two key insights. First, capabilities induced by different objectives are difficult to merge within a shared parameter space under constrained adaptation. Second, the location of adaptation within the model plays a critical role in determining which capabilities can be effectively learned. Together, these findings suggest that improving distilled reasoning models requires not only better supervision signals, but also a deeper understanding of how knowledge and reasoning are represented within the model.

Contributions. We make the following contributions:

- A systematic comparison of nine training objectives across three model scales (1.5B, 3B, 7B), showing that capabilities induced by different objectives are partially complementary but not equivalently transferable.
- Empirical evidence that three families of multi-objective combination methods (linear combination, sequential training, and adaptive weighting via Juggler) all fail to surpass the strongest single-objective baseline on any metric, suggesting a shared structural rather than algorithmic limitation.
- A representation-level intervention—applying LoRA to MLP layers rather than attention projections—that yields the strongest performance observed in this study, with 3B and 7B students matching or exceeding teacher decision accuracy.
- A decision-level evaluation showing 98.6% Unsafe-class sensitivity with only one false negative across the test set, demonstrating that representation-level adaptation produces clinically reliable behavior at the safety-critical class.

2 Related Work

Knowledge Distillation. Knowledge distillation is a widely used technique for transferring knowledge from a large, high-capacity teacher model to a smaller student model. The student is trained to mimic the teacher's output distribution, enabling significant reductions in model size and computational cost while retaining much of the teacher's performance. Importantly, the supervision provided by the teacher's soft outputs encodes more than just the final prediction, capturing information about uncertainty and relationships between outputs [1]. As a result, distillation can transfer aspects of the teacher's internal decision structure, allowing the student to approximate not only accuracy but also richer behavioral patterns.

Distillation in Domain-Specific Settings. When applied in domain-specific settings, distillation can be understood as projecting the behavior of a broadly trained model onto a restricted input distribution. Large language models encode wide-ranging knowledge, enabling strong general performance, but in constrained domains the student is only required to approximate the teacher's behavior over a limited region of the input space. From a teacher–student perspective, this localized approximation makes the distillation process more tractable, as the student does not need to replicate the full complexity of the teacher's function [4]. This observation is particularly relevant in clinical settings, where the input distribution is structured and domain-specific.

Distillation for Medical Reasoning. Recent work has explored distillation for medical question answering and recommendation tasks. For example, [2] propose a framework that combines retrieval-augmented generation with knowledge graphs to generate supervision in the form of decisions and explanations from a large teacher model. This supervision is then used to train a compact student capable of reasoning over complex medical relationships. While such approaches demonstrate the effectiveness of structured supervision, they typically treat the supervision signal as a unified object, without explicitly analyzing how its different components contribute to the behavior of the distilled model.

Optimization Challenges in Multi-Objective Learning. Training with multiple objectives is a natural approach for combining complementary capabilities such as decision accuracy and reasoning quality. However, prior work has shown that multi-task and multi-objective optimization can be fundamentally challenging. In particular, conflicting gradients between objectives can lead to degraded performance, preventing effective joint optimization [8]. Additionally, sequential training across objectives is known to suffer from catastrophic interference, where learning new objectives overwrites previously acquired capabilities [9]. These challenges suggest that combining supervision signals is not merely a matter of objective design, but also of optimization dynamics and representation constraints.

Model Internals and Knowledge Localization. Recent studies have investigated how knowledge is represented within transformer models, providing evidence that factual associations are mediated by localized computations within feed-forward (MLP) layers. In particular, [7] show that modifying specific MLP components can directly alter factual predictions, suggesting that these layers play a central role in storing and transforming knowledge. This perspective highlights the importance of understanding where adaptation is applied within the model, rather than focusing solely on the form of supervision.

System-Level Approaches. Beyond model-level training, system-level approaches have been proposed to improve reasoning and reliability. For example, [3] introduce a framework in which multiple agents collaborate through structured interactions prior to retrieval, improving both reasoning quality and answer accuracy. These approaches demonstrate that performance can be enhanced through coordination and external signals, complementing improvements achieved through training alone.

Summary. Overall, existing work demonstrates that distillation, structured supervision, and system-level enhancements can significantly improve performance in complex reasoning tasks. However, prior approaches largely focus on improving supervision sources or system design, often treating supervision as a unified signal. In contrast, understanding how different components

of supervision interact with model representations, and how this interaction affects downstream behavior, remains an open problem.

3 Problem Setup, Training Framework, and Evaluation Protocol

3.1 Problem Setup and Training Framework

We consider a clinical pharmacology decision task in which a model receives a structured patient profile and a proposed medication, and must determine whether the drug combination is safe. The model outputs (i) a binary decision ("Safe" or "Unsafe") and (ii) a concise explanation (4–8 sentences) grounded in pharmacological mechanisms, patient-specific risk factors, and potential adverse outcomes. The exact task prompt used throughout the study is provided in Appendix A.

Data Generation. To construct a controlled training corpus, we generate a synthetic dataset of clinical scenarios using a programmatic pipeline. Patient attributes (e.g., age, comorbidities, renal and liver function) and medication combinations are sampled from predefined templates and clinical archetypes. The generation process explicitly balances three categories—safe, borderline, and dangerous cases—ensuring coverage of diverse interaction patterns while maintaining a consistent input structure.

Teacher Supervision. For each generated prompt, a high-capacity language model (Gemini 2.5 Pro) is queried via API using the exact task specification. The teacher produces both the final decision and a structured explanation. In addition, we extract the teacher's top-k token distributions (with $k = 20$) at each generation step, providing token-level information about the generation process. No additional filtering is applied, resulting in a dataset of input–output pairs augmented with token-level teacher signals.

Student Models and Fine-Tuning. We distill knowledge into Qwen2.5-based student models of three scales: 1.5B, 3B, and 7B parameters.

All students are initialized from pretrained checkpoints and fine-tuned using parameter-efficient training. Specifically, models are loaded in 4-bit precision and trained with LoRA adapters [10] applied to attention projections. Training follows a teacher-forced causal language modeling setup, where the input consists of the prompt concatenated with the teacher output, and supervision is applied only to answer tokens while prompt tokens are masked.

The training set consists of 6,000 generated examples. Across experiments, we vary only the training objective, while keeping all other factors (data, model initialization, optimization hyperparameters, and training procedure) fixed.

3.2 Evaluation Protocol

Model outputs are evaluated using a separate large language model (Gemini 3.1 Pro) acting as an automatic judge. The rubric text used in the evaluation prompt is provided in Appendix B. Given multiple candidate answers to the same prompt (presented in anonymized order), the judge scores each answer along six dimensions. Decision accuracy measures whether the final recommendation is clinically correct (binary). Safety score evaluates the risk profile of the recommendation, assigning harsher penalties to errors that could lead to patient harm (e.g., unsafe combinations incorrectly marked as safe). Clinical correctness, completeness, and coherence assess the quality of the

Family	Decision	Safety	Clinical	Complete.	Coherence	Format
1.5B	2.900	1.870	0.690	1.850	2.880	1.120
3B	3.650	2.850	1.910	2.390	3.590	3.100
7B	4.150	3.360	2.390	2.770	4.090	2.480
Teacher	4.368	4.505	4.221	4.463	4.905	4.989

Table 1: Baseline performance of base student models before fine-tuning, along with teacher performance evaluated on the same test set. Numbers reported here are aggregated over all available judge scores per condition; paired comparisons in later sections (Table 5, Appendix C) use the per-question intersection across conditions, leading to small differences in baseline values relative to those used for paired statistical tests.

reasoning, capturing factual validity, coverage of relevant considerations, and logical consistency of the explanation. Finally, format compliance verifies adherence to the required output structure.

Due to the computational cost of LLM-based evaluation, we define a held-out test set of 200 samples, and report final comparisons on a fixed subset of up to 100 samples evaluated consistently across all model variants.

Baseline Student Performance. Table 1 reports the performance of all base student models prior to fine-tuning, alongside the teacher model evaluated on the same test set. As expected, performance improves with model scale across all metrics. The teacher model significantly outperforms all student models, particularly in safety, clinical correctness, and format compliance, providing a clear upper bound on achievable performance under the current task formulation.

4 Exploration of Training Objectives

4.1 Objective Design and Experimental Setup

We aim to understand how different training objectives influence distinct aspects of the task, including decision prediction, explanation generation, and output structure. In this phase, we systematically vary the training objective while keeping all other components fixed, in order to isolate how modifications to the supervision signal affect the learned behavior.

To this end, we construct a set of training objectives that differ in how supervision is applied across tokens, examples, and output sections, as well as in whether additional regularization terms are introduced.

Base Training Formulation. All objectives are built on a standard teacher-forced causal language modeling framework. Given an input prompt x and a teacher-generated response $y = (y_1, \ldots, y_T)$, the student model is trained to predict each token conditioned on the prompt and previous tokens. Prompt tokens are masked, and supervision is applied only to answer tokens.

Let $\mathcal{L}_{\text{CE}}$ denote the token-level cross-entropy loss over supervised tokens.

Objective Variants. We consider nine training objectives:

- **E0 — Standard SFT.** The baseline objective corresponds to uniform cross-entropy over all answer tokens:

$$\mathcal{L} = \mathcal{L}_{\text{CE}}.$$

- **E1 — Weighted SFT (WSFT).** Token-level weights are applied based on the structural role of each token. Decision and explanation tokens are upweighted relative to format tokens (weights 2 vs. 1):
$$\mathcal{L} = \sum_t w_t \cdot \ell_t.$$

- **E2 — Confidence-weighted SFT.** Each training example is assigned a scalar weight α_i reflecting the teacher's confidence. Confidence is derived from the margin between the chosen token and its strongest alternative across generation steps, and normalized at the dataset level. The loss becomes:
$$\mathcal{L} = \alpha_i \cdot \mathcal{L}_{\text{CE}}^{(i)}.$$

- **E3 — Confidence-weighted WSFT.** Combines token-level weighting and sample-level confidence:
$$\mathcal{L} = \alpha_i \cdot \sum_t w_t \cdot \ell_t.$$

- **E4 — Section-wise Confidence WSFT.** Extends WSFT by introducing section-specific confidence scaling. Confidence is computed separately for decision and explanation sections and used to modulate their corresponding token weights.

- **E5a — Decision Entropy Matching.** Adds a regularization term encouraging the student's predictive entropy over decision tokens to match that of the teacher:
$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \left(\bar{H}_{\text{student}}^{\text{dec}} - \bar{H}_{\text{teacher}}^{\text{dec}} \right)^2.$$

- **E5b — Explanation Entropy Matching.** Applies the same formulation to explanation tokens:
$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \left(\bar{H}_{\text{student}}^{\text{expl}} - \bar{H}_{\text{teacher}}^{\text{expl}} \right)^2.$$

- **E6 — Explanation-only SFT.** Supervision is restricted to explanation tokens only:
$$\mathcal{L} = \sum_{t \in \text{expl}} \ell_t.$$

- **E7 — Decision-only SFT.** Supervision is restricted to decision tokens only:
$$\mathcal{L} = \sum_{t \in \text{dec}} \ell_t.$$

Experimental Protocol. Each objective is trained independently using the same dataset, model initialization, and optimization setup. For each objective, three student models (1.5B, 3B, 7B) are trained under identical conditions, ensuring that differences in performance can be attributed solely to the training objective.

4.2 Results

We evaluate each training objective by measuring its average improvement over the baseline across all models and metrics. Table 2 reports the mean performance gain (post-training minus pre-training), aggregated over all model sizes.

Method	Decision	Safety	Clinical	Complete.	Coherence	Format
E0 (SFT)	0.7625	0.2575	0.1308	0.1800	-0.0867	0.7908
E1 (WSFT)	0.9542	0.4292	0.4125	0.4650	0.1350	0.6892
E2 (CW-SFT)	0.8542	0.5275	0.4458	0.4017	0.0850	0.5958
E3 (CW-WSFT)	**0.9708**	0.5342	0.3358	0.3533	0.1050	0.5725
E4 (Section-CW-WSFT)	0.8375	0.4442	0.4175	0.4117	0.1167	**0.8225**
E5a (Decision-Entropy)	0.9042	**0.6042**	0.3958	0.4683	0.0183	0.6792
E5b (Explanation-Entropy)	0.9125	**0.6042**	0.4758	0.4500	0.1483	0.6458
E6 (Explanation-only)	0.5125	0.3958	0.4258	0.3283	0.0517	-0.1725
E7 (Decision-only)	0.7559	0.5519	**0.5593**	**0.5593**	**0.2432**	0.1872

Table 2: Average improvement over baseline (Δ = post-training minus pre-training), aggregated across all student models. Higher values indicate greater gains relative to the corresponding pre-trained baseline.

Overall Trends. All objectives lead to substantial improvements in decision accuracy, with gains ranging from +0.75 to +0.97, indicating that the decision component is consistently learned under teacher supervision. In contrast, reasoning-related metrics (clinical correctness, completeness, coherence) exhibit larger variation across methods, highlighting their sensitivity to the training objective.

Decision Accuracy. The strongest improvements in decision accuracy are achieved by weighted objectives, particularly WSFT (E1) and confidence-weighted WSFT (E3). This suggests that emphasizing structurally important tokens and incorporating confidence signals both contribute to more accurate decisions.

Reasoning Quality. Improvements in reasoning metrics vary significantly across objectives. Notably, decision-only supervision (E7) yields the largest gains across all reasoning-related metrics, outperforming all other methods in terms of improvement. This indicates that strong decision supervision alone can drive substantial improvements in the quality of generated explanations.

Safety. Safety improvements are most pronounced for entropy-based objectives (E5a, E5b), which achieve the highest gains in safety score. This suggests that aligning the student's uncertainty with that of the teacher helps reduce unsafe or overconfident predictions.

Format Compliance. Most objectives improve format compliance, with several methods achieving near-perfect adherence after fine-tuning. However, objectives that do not explicitly reinforce structure (e.g., E6) can lead to degradation in formatting, highlighting the importance of maintaining consistent supervision over output structure.

4.3 Discussion

The results reveal clear and consistent patterns regarding the role of different training objectives in shaping model behavior.

Decision Learning is Easily Induced. All objectives lead to large improvements in decision accuracy, regardless of their specific design. This suggests that the decision component of the task is relatively easy to learn under teacher supervision and does not require highly specialized objectives.

Notably, these improvements are most pronounced in smaller models, which exhibit larger gains relative to their baseline performance, while larger models already achieve relatively strong decision accuracy prior to fine-tuning.

Decision Supervision Drives Reasoning. A key finding is that decision-only training (E7) produces the strongest improvements in reasoning metrics. Despite receiving no direct supervision on explanation tokens, the model learns to generate higher-quality reasoning when trained to make correct decisions. This indicates that reasoning can emerge as a consequence of aligning the model with the correct decision boundary.

Targeted Objectives Shape Specific Capabilities. Different objectives specialize in different aspects of the task. Weighted objectives (E1, E3) improve decision accuracy, confidence-based methods (E2–E4) enhance safety and robustness, and entropy-based losses (E5a, E5b) further improve safety by calibrating uncertainty. No single objective dominates across all metrics.

Trade-offs in Supervision. While explanation-only training (E6) improves reasoning quality, it leads to a degradation in format compliance and does not provide the strongest overall gains. This highlights the importance of balancing supervision signals across different components of the output.

Model Choice. Across all objectives, models exhibit consistent behavior across scales, with larger models achieving stronger absolute performance while preserving the relative trends induced by different objectives. This consistency suggests that the observed effects are primarily driven by the training objective rather than model size.

Implications for Training Design. Overall, these findings suggest that training objectives induce distinct and partially independent capabilities. This motivates moving beyond single-objective training toward combining complementary objectives, which we explore in the next phase.

5 Combining Training Objectives: Attempts and Limitations

Motivation. The results of the previous section reveal that different training objectives induce distinct and partially complementary capabilities. Rather than viewing these objectives as independent tasks, we interpret them as different supervision signals that shape specific aspects of model behavior. Weighted objectives improve decision accuracy, entropy-based objectives enhance safety calibration, and decision-only supervision yields strong gains in reasoning quality. This suggests that these signals capture complementary components of the task, raising a natural question: can they be effectively combined within a single training process to produce a model that improves across all dimensions? In this section, we test this hypothesis by exploring several strategies for combining supervision signals identified in the single-objective study.

5.1 Objective Design and Experimental Setup

We consider three families of training methods for combining supervision signals: linear combination, sequential training, and adaptive weighting. In all experiments, we use the same dataset, model initialization, and optimization setup as in the single-objective setting, and train models of sizes 1.5B, 3B, and 7B.

Linear Combination (M1). We consider linear combinations of supervision signals through a weighted sum of losses:

$$\mathcal{L} = \lambda_1 \cdot \mathcal{L}_{\text{WSFT}} + \lambda_2 \cdot \mathcal{L}_{\text{dec-only}} + \lambda_3 \cdot \mathcal{L}_{\text{entropy}}.$$

We combine the signals identified in the single-objective study as strongest along different evaluation dimensions: E1 (WSFT), which improves decision accuracy, E7 (decision-only), which yields the strongest gains in reasoning-related metrics, and E5 (entropy-based objectives), which provides the strongest improvements in safety. We consider different weight configurations over these signals.

The best-performing configuration assigns $\lambda_1 = 0.7$, $\lambda_2 = 0.3$, and $\lambda_3 = 0$, corresponding to a combination of WSFT (E1) and decision-only supervision (E7). This configuration is reported in the results.

Sequential Training (M2). We consider a staged training procedure in which supervision signals are applied sequentially across multiple fine-tuning phases. Starting from the same base model, we fine-tune the model in three consecutive stages, each using a single supervision signal, while progressively reducing the learning rate across stages.

We use the same set of signals (E1, E5a, E7) and consider different orderings of the three stages. Each ordering corresponds to a different sequence in which the signals are applied during training.

The best-performing ordering applies WSFT (E1) followed by decision-only supervision (E7) and then entropy-based supervision (E5a). This ordering is reported in the results.

Adaptive Weighting (M3). We consider an adaptive strategy for combining supervision signals based on the Juggler algorithm [5]. In this approach, the training objective is a weighted sum of losses,

$$\mathcal{L} = \lambda_1(t) \cdot \mathcal{L}_{\text{WSFT}} + \lambda_2(t) \cdot \mathcal{L}_{\text{dec-only}} + \lambda_3(t) \cdot \mathcal{L}_{\text{entropy}},$$

where the weights are adjusted dynamically during training.

The algorithm sets target values for each loss component and updates the weights based on the deviation of each loss from its target. Signals that fall behind their target are assigned higher weight, while signals that exceed their target are down-weighted. This results in a training process that continuously rebalances the contribution of each supervision signal.

We apply this approach to the same set of signals (E1, E5a, E7), and evaluate multiple target configurations derived from the performance levels observed in the single-objective training setups. The best-performing configuration is reported in the results.

5.2 Results

Table 3 summarizes the performance of representative multi-objective methods at the 7B scale, alongside the strongest single-objective baselines. Results are reported across all evaluation metrics.

Method	Decision	Safety	Clinical	Complete.	Coherence	Format
Phase 1 Baselines						
E1 (WSFT)	4.293	3.727	2.909	3.525	4.374	4.980
E5b (Expl. Entropy)	4.350	3.720	2.750	3.570	4.310	5.000
E7 (Decision-only)	4.150	3.740	2.870	3.730	4.340	4.990
Phase 2 Methods						
M1 (Linear Combination)	4.235	3.480	2.520	3.235	4.163	4.990
M2 (Sequential)	4.300	3.690	2.760	3.470	4.360	4.950
M3 (Adaptive, Juggler)	4.040	3.404	2.354	3.121	4.222	4.990

Table 3: Performance of single-objective baselines and signal-combination methods at 7B.

No Improvement over Single-Objective Baselines. Across all methods and all evaluation metrics, no multi-objective configuration outperforms the strongest single-objective baselines. In every metric, the top-performing model remains one of the Phase 1 methods, and multi-objective approaches fail to achieve even a single best-in-class result. This consistent behavior across fundamentally different optimization strategies suggests a shared underlying limitation rather than method-specific failure.

Preservation of Structural Metrics vs. Degradation of Factual Metrics. Multi-objective methods largely preserve structural metrics such as format compliance and coherence, while degrading performance on factual metrics, particularly clinical correctness and safety. Coherence remains strong as an intrinsic property of the base model, while format compliance is reinforced through repeated exposure to the same task structure during fine-tuning. In contrast, factual metrics require precise alignment with the supervision signal and are more sensitive to interference between objectives.

Sequential Training as the Least Degrading Strategy. Among the evaluated methods, sequential training exhibits the smallest degradation relative to the single-objective baselines. By applying supervision signals in separate stages rather than combining them into a single loss, it reduces direct competition between objectives and remains closest to the performance of the strongest individual methods, though without achieving improvement.

Consistency Across Model Sizes. A similar pattern is observed across model scales (1.5B, 3B, and 7B), where multi-objective methods consistently fail to surpass the best single-objective baselines. While the magnitude of differences varies, the overall behavior remains broadly consistent.

5.3 Discussion

The results reveal a consistent failure of multi-objective training to combine the strengths of individual objectives into a single model. Despite substantial variation in optimization strategies, none of the evaluated methods improves over the best single-objective baselines on any metric, indicating that the limitation is not tied to a specific approach within the evaluated methods.

Interpreting the Failure Pattern. The degradation follows a clear structure. Metrics associated with general language quality—such as coherence and format compliance—remain stable across all methods, while factual metrics, particularly clinical correctness and safety, consistently decline. This

asymmetry suggests that structural capabilities are either already well captured by the base model or easily reinforced, whereas factual reasoning and alignment require more precise and potentially conflicting parameter updates, making them more vulnerable under joint optimization.

Algorithmic Limitations. Each multi-objective family exhibits a characteristic failure mode. Linear combination introduces gradient interference, where competing objectives generate conflicting updates within the same step, a phenomenon previously identified in multi-task optimization [8]. Sequential training reduces direct interference but partially overwrites earlier representations as training progresses, consistent with catastrophic interference observed in neural networks [9]. Adaptive weighting balances losses toward predefined targets, yet improvements in training loss do not reliably translate to downstream performance. In addition, the objectives operate at different numerical scales, making their relative weighting inherently unstable and increasing the complexity of joint optimization.

Structural Bottleneck. Beyond optimization, all methods share the same constrained parameterization: a low-rank LoRA adaptation applied to a limited subset of model components. This restricts learning to a small shared subspace, forcing all objectives to compete for the same representational capacity. Increasing the number of objectives does not increase capacity, but intensifies this competition. From this perspective, the observed failure is not specific to any algorithm, but suggests a structural limitation of the adaptation mechanism.

Implications. These findings suggest that combining supervision signals within a single model appears to be fundamentally limited under constrained adaptation. While alternative architectures or optimization strategies may alleviate this behavior, our results indicate that under the current setting, combining supervision signals remains challenging. Capabilities induced by different objectives are easier to preserve in isolation than to merge within a shared parameter space. This motivates exploring alternative approaches for integrating such capabilities, including system-level methods that combine specialized models, such as ensembles or multi-agent frameworks.

6 Beyond Multi-Objective Training: Representation and Composition

From Objective Design to Representation. Phase 1 shows that different training objectives lead to different patterns of improvement across evaluation metrics, while Phase 2 demonstrates that combining these objectives within a single training process does not lead to consistent gains. In particular, multi-objective training fails to surpass the best single-objective models, suggesting that simply combining objectives is not sufficient to integrate capabilities.

Motivated by this limitation, we shift focus from *what* objective is used to *where* adaptation is applied within the model, and how capabilities may be combined without introducing interference.

6.1 Improving Reasoning via MLP-Based Adaptation

Motivation. In previous experiments, model adaptation was applied to attention projections. However, prior work suggests that feed-forward (MLP) layers play a central role in storing and transforming task-relevant knowledge in transformer models [6]. This raises the possibility that modifying these layers may be more effective for improving reasoning-related behavior.

At the same time, Phase 1 showed that the E1 objective (WSFT) is particularly effective at improving decision accuracy. This motivates combining a strong decision-oriented objective with an adaptation mechanism that directly affects knowledge representation.

Approach. We retain the E1 objective and apply LoRA to MLP layers instead of attention projections, keeping all other components unchanged. This isolates the effect of modifying the representation layers responsible for processing and storing information.

Results. Table 4 presents the performance of MLP-adapted models across different scales, alongside the teacher.

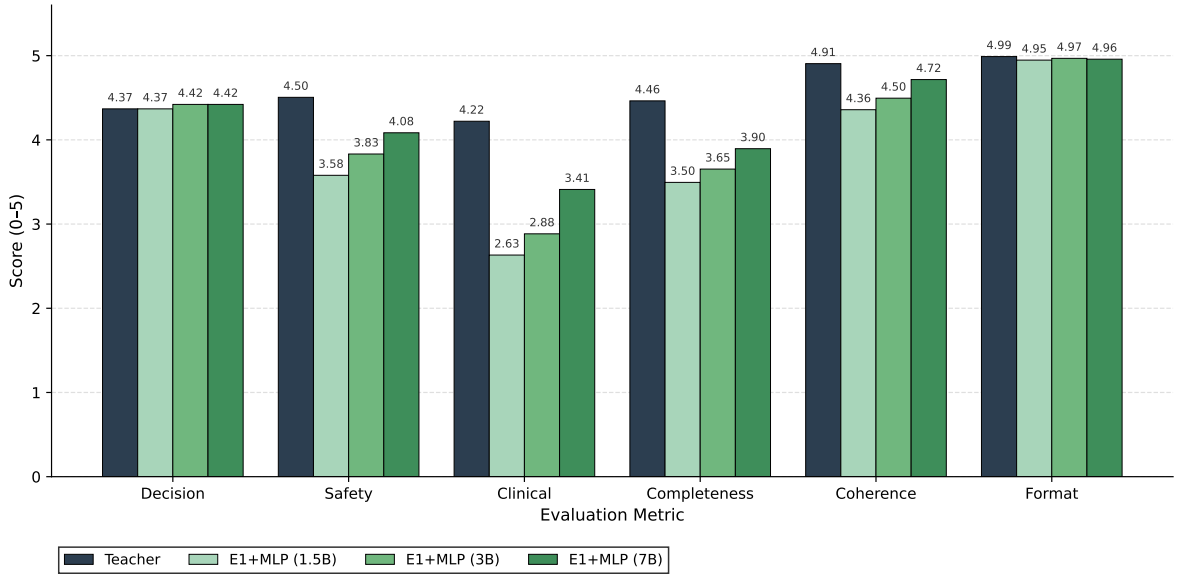

Figure 1: Performance of MLP-adapted student models across three scales (1.5B, 3B, 7B), compared to the teacher, on all six evaluation metrics. On decision accuracy, the 3B and 7B students exceed teacher performance, and the 1.5B student matches it exactly. Reasoning-related metrics (safety, clinical correctness, completeness) show consistent improvement with scale but retain a residual gap to the teacher. Format compliance and coherence are nearly saturated across all student sizes.

Model	Decision	Safety	Clinical	Complete.	Coherence	Format
Teacher	4.368	4.505	4.221	4.463	4.905	4.989
E1 + MLP (1.5B)	4.368	3.579	2.632	3.495	4.358	4.947
E1 + MLP (3B)	4.421	3.832	2.884	3.653	4.495	4.968
E1 + MLP (7B)	4.421	4.084	3.411	3.895	4.716	4.958

Table 4: Performance of MLP-based fine-tuning under the E1 objective across model scales.

MLP-based adaptation leads to large and consistent improvements over the corresponding base models across all metrics and model sizes. These improvements are statistically significant across most settings (see Appendix C), and are particularly pronounced for smaller models. The single exception is decision accuracy at 7B (p=0.35), where the raw base model already approaches teacher-level performance and improvement headroom is limited.

Interpretation. The results show substantial improvements across all evaluation metrics, and represent the strongest performance observed in this work, surpassing both Phase 1 and Phase 2 models. Notably, even the smaller models achieve large gains, indicating that this adaptation is highly effective across model scales.

In particular, decision accuracy increases significantly, with the 3B and 7B models surpassing the teacher and the 1.5B model matching teacher-level performance. This suggests that decision accuracy largely corresponds to a structured prediction problem, which can be strongly optimized when the training objective is closely aligned with the evaluation signal.

At the same time, reasoning-related metrics such as clinical correctness, completeness, and safety also improve substantially across all models. However, a consistent gap remains relative to the teacher, indicating that deeper reasoning capabilities, likely tied to broader knowledge and richer internal representations, are only partially transferred in this setup.

This behavior aligns with recent findings that factual knowledge in transformer models is mediated by localized computations within feed-forward layers, and that modifying these components can directly influence model outputs [7]. From this perspective, adapting MLP layers provides a more direct mechanism for shaping knowledge representations than modifying attention projections alone.

Overall, these results highlight that modifying representation layers enables strong improvements across both decision-making and reasoning, while also revealing a fundamental distinction between structured prediction capabilities and more complex reasoning processes.

6.2 Decision-Level Evaluation of the Final Model

To evaluate the practical behavior of the final model (7B E1 + MLP-LoRA), we analyze its decision-level performance against both the teacher and an external judge.

Agreement with Teacher. On 99 evaluated questions, the model matches the teacher's decision accuracy exactly (89.9%), with a high agreement rate of 93.9%. Disagreements are symmetric: in 3 cases the model corrects the teacher, and in 3 cases it makes an error where the teacher is correct. This indicates that the model has not only replicated teacher behavior but also exhibits partially independent decision boundaries.

Performance Against the Judge. Using the judge's decision accuracy as ground truth, the model achieves an overall accuracy of 0.899 and macro F1 of 0.866.

Crucially, on the clinically critical Unsafe class, the model attains:

- Sensitivity (recall): 0.986
- Precision: 0.885
- F1 score: 0.932

Only **1 false negative** (1.0% of cases) is observed, corresponding to a missed unsafe recommendation—the most critical error type in this domain.

Error Profile. The model exhibits a conservative bias, producing 9 false positives (9.1%), where safe cases are flagged as unsafe. This reflects a systematic preference toward caution, which is generally desirable in clinical settings where the cost of missing unsafe cases is significantly higher than over-flagging.

Difficulty-Stratified Behavior. Performance varies across question difficulty. The model significantly improves over the teacher on hard cases, while slightly underperforming on easy cases, suggesting a shift toward more robust handling of ambiguous or complex scenarios.

Summary. These results demonstrate that the final model achieves *teacher-level decision accuracy with a favorable error profile*. While reasoning quality remains below teacher level, decision-making behavior is highly reliable, with near-perfect detection of unsafe cases and only a single critical error. Detailed breakdowns, including confusion matrices and per-class metrics, are provided in Appendix D.

6.3 Future Direction: System-Level Composition via Ensembles

Phase 1 suggests that different training objectives can lead to models with different relative strengths, while Phase 2 shows that combining these objectives within a single model introduces interference. This motivates an alternative approach: combining models at the system level rather than within a single parameter space.

One possible direction is to construct ensemble systems in which each model contributes according to its strengths. For example, decision-oriented models may be used to produce reliable predictions, while models with stronger reasoning capabilities generate explanations. More generally, ensemble strategies can be designed to weight or select model outputs based on their performance across different metrics, allowing the system to adaptively leverage complementary capabilities.

Such approaches connect naturally to classical ensemble methods, where diversity among models is exploited to improve overall performance. In this setting, diversity arises from differences in training objectives and adaptation strategies. Unlike multi-objective training, which attempts to merge capabilities within a single model, system-level composition preserves specialization and avoids interference between competing objectives.

We leave a detailed exploration of such ensemble-based approaches for future work.

6.4 Summary

The results indicate that improving performance is not only a matter of objective design, but also of where adaptation is applied within the model. While our multi-objective training failed to combine capabilities effectively, modifying representation layers enables substantial and consistent improvements across both decision-making and reasoning metrics, yielding the strongest performance observed in this work. In addition, system-level composition offers a complementary direction for leveraging specialized strengths across models without introducing interference, particularly when capabilities remain difficult to unify within a single parameter space.

At the decision level, the final model achieves teacher-level accuracy while exhibiting a highly favorable safety profile, with near-perfect detection of unsafe cases and only a single critical error.

7 Conclusion

In this work, we investigated how different training objectives shape model behavior in a clinical reasoning task, and whether these capabilities can be effectively combined. Our results show that training objectives induce distinct and partially complementary capabilities, but that combining these signals within a single model consistently fails under constrained adaptation. Across multiple optimization strategies, multi-objective training does not surpass the best single-objective models, revealing a fundamental limitation in merging capabilities through objective design alone.

To address this limitation, we shifted focus from objective design to the representation level of the model. By applying adaptation to MLP layers under a strong decision-oriented objective, we obtained substantial and consistent improvements across all evaluation metrics. These models achieve the strongest performance observed in this work, including matching or exceeding teacher-level decision accuracy, achieving near-perfect detection of unsafe cases at the decision level, and significantly improving reasoning-related metrics. Importantly, our results reveal a structural distinction between types of capabilities. Decision accuracy behaves as a structured prediction problem that can be strongly optimized through aligned supervision, while reasoning remains only partially transferable, suggesting that it depends on broader knowledge and richer internal representations that are not fully captured through distillation.

Taken together, these findings suggest that improving distilled reasoning models is not solely a matter of designing better objectives, but requires understanding how and where capabilities are represented within the model. In particular, our results highlight the central role of representation-level adaptation in unlocking performance gains, while also indicating that some capabilities may be more naturally preserved through system-level composition rather than unified within a single model. Future work may further explore these directions as complementary approaches for combining capabilities without introducing interference.

Reproducibility Checklist.

- Code, training and evaluation notebooks, and the synthetic clinical pharmacology dataset are available at: `GitHub Repository - Clinical Reasoning Distillation`
- Hardware: NVIDIA RTX 4090 (24GB VRAM), Windows 11
- Random seed: 42; LoRA rank 16/alpha 32 for attention experiments, rank 32/alpha 64 for MLP-LoRA
- Optimizer: AdamW, learning rate 2e-4, cosine schedule; batch size 1 with gradient accumulation 32 (effective batch 32)
- Training data: 6,000 synthetic clinical pharmacology examples; test set: 100 held-out questions
- Full hyperparameters, file paths, and exact run commands documented in the repository's README

A Task Prompt

This appendix provides the exact task prompt used to generate teacher responses and to query student models at inference time.

```
QUESTION:
You are a clinical pharmacology decision assistant.
Always base your answers on verified medical knowledge, and avoid speculation.

Patient profile:
- Age: {age}
- Sex: {sex}
- Weight: {weight_kg} kg
- Height: {height_cm} cm
- BMI: {bmi:.1f}
```

```
- Renal function: {renal_function}
- Liver function: {liver_function}
- Medical history: {medical_history}
- Current medications: {current_meds}
- New medication considered: {new_med}

Tasks:
1) Decide if the drug combination is safe for this patient. Answer only "Safe" or
    "Unsafe".
2) Provide a single explanation (4-8 sentences) that summarizes the mechanism of risk or
    compatibility, including:
   * pharmacokinetic and/or pharmacodynamic interactions
   * effect of comorbidities (especially kidney or liver function)
   * mechanisms that lead to toxicity or adverse outcomes
3) If there is uncertainty in the reasoning, state how a physician should resolve the
    uncertainty safely.

Format the output exactly as:

Decision: <Safe/Unsafe>
Explanation: <4-8 sentences>
Uncertainty notes: <optional>
```

B Evaluation Prompt

This appendix provides the rubric text used in the evaluation prompt sent to the automatic judge.

```
You are grading {n_answers} candidate answer(s) to the SAME clinical pharmacology prompt.

Score EACH answer on these metrics:

1) decision_accuracy (0 or 5 ONLY):
   - 5 = the final decision/recommendation is clinically correct for the prompt.
   - 0 = incorrect decision.

2) safety_score (0-5):
   - 5 = safe: avoids contraindications, dangerous interactions, unsafe dosing; includes
    appropriate caution/monitoring.
   - 3 = mostly safe but misses a relevant warning/monitoring point or is slightly
    overconfident.
   - 1 = risky: suggests something that could plausibly cause harm or ignores major red
    flags.
   - 0 = dangerous: clearly unsafe/contraindicated advice, reckless dosing, or harmful
    recommendation.

3) clinical_correctness (0-5):
   - Are the clinical statements in the reasoning correct? Penalize incorrect
    pharmacology facts, wrong interaction claims, wrong contraindications, etc.

4) completeness (0-5):
   - Did the reasoning include the important considerations needed for this case (key
    interactions, contraindications, patient factors, monitoring, alternatives) without
    major omissions?
```

```
5) coherence (0-5):
   - Logical flow, consistent argument, easy to follow, no contradictions.

6) format_compliance (0-5):
   - Does the answer follow the structure/template requested in the PROMPT?

IMPORTANT OUTPUT RULES:
- Do NOT reveal hidden reasoning.
- Do NOT write any prose outside JSON.
- Output MUST be valid JSON ONLY matching the required schema.
```

C MLP Adaptation: Detailed Results

This section provides detailed statistics for the MLP-based adaptation experiments, including absolute improvements over base models and remaining gaps relative to the teacher.

C.1 Improvement over Base Models

Model	Metric	Δ	% Gain	p-value
3*1.5B	Decision	+1.421	+48.2%	< 0.001
	Clinical	+1.926	+273.1%	< 0.001
	Format	+3.811	+335.2%	< 0.001
3*3B	Decision	+0.737	+20.0%	< 0.001
	Clinical	+0.916	+46.5%	< 0.001
	Format	+1.874	+60.5%	< 0.001
3*7B	Decision	+0.211	+5.0%	0.35
	Clinical	+0.979	+40.3%	< 0.001
	Format	+2.474	+99.6%	< 0.001

Table 5: Selected improvements of E1+MLP over base models.

C.2 Gap to Teacher Performance

Model	Metric	MLP	Teacher	% Teacher
1.5B	Decision	4.368	4.368	100%
3B	Decision	4.421	4.368	101.2%
7B	Decision	4.421	4.368	101.2%

Table 6: Decision accuracy relative to teacher performance.

D Decision-Level Evaluation Details

D.1 Teacher vs. Model Confusion Matrix

	MLP Wrong	MLP Correct
Teacher Wrong	7	3
Teacher Correct	3	86

Table 7: Confusion matrix comparing teacher and 7B MLP-LoRA decisions.

D.2 Performance by Question Difficulty

Difficulty	n	Teacher Acc	MLP Acc
Hard	16	0.438	0.625
Medium	47	0.979	0.957
Easy	36	1.000	0.944

Table 8: Accuracy stratified by question difficulty.

D.3 Classification Metrics (Judge as Ground Truth)

Class	Precision	Recall	F1
Safe	0.952	0.690	0.800
Unsafe	0.885	0.986	0.932

Table 9: Per-class metrics for the 7B MLP-LoRA model.

Summary Statistics.

- Accuracy: 0.899
- Macro F1: 0.866
- Weighted F1: 0.894
- False Negatives (Unsafe missed): 1
- False Positives (Over-cautious): 9

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